1 LASSO when $X^\top X = I$

Consider the case $X^\top X = I$.

In this case of LASSO, we minimize

$$\Delta = \sum_{i=1}^{n}\Big(y_i - \sum_{j=1}^{p}\beta_j x_{ji}\Big)^2 + \lambda\sum_{j=1}^{p}|\beta_j|. \tag{1}$$

Case $\beta_k > 0$

If $\beta_k > 0$,

$$\frac{\partial\Delta}{\partial\beta_k} = \sum_{i=1}^{n}\Big\{2\Big(y_i - \sum_{j=1}^{p}\beta_j x_{ji}\Big)(-x_{ki})\Big\} + \lambda.$$

Setting the derivative to zero:

$$\frac{\partial\Delta}{\partial\beta_k} = 0 \quad\Rightarrow\quad -\sum_{i=1}^{n} y_i x_{ki} + \sum_{i=1}^{n}\sum_{j=1}^{p}\beta_j x_{ji}x_{ki} + \frac{\lambda}{2} = 0.$$

Expanding the second summation,

$$-\hat\beta_{k,\mathrm{LS}} + \beta_k\sum_{i=1}^{n}x_{ki}^2 + \beta_2\sum_{i=1}^{n}x_{2i}x_{ki} + \cdots + \beta_p\sum_{i=1}^{n}x_{pi}x_{ki} + \frac{\lambda}{2} = 0.$$

Since $X^\top X = I$, all cross-product terms vanish and $\sum_{i=1}^{n} x_{ki}^2 = 1$. Hence,

$$\hat\beta_{k,\mathrm{LS}} = \beta_k + \frac{\lambda}{2} \quad\Rightarrow\quad \beta_k = \hat\beta_{k,\mathrm{LS}} - \frac{\lambda}{2}.$$

$$\boxed{\hat\beta_{k,\mathrm{LASSO}} = \hat\beta_{k,\mathrm{LS}} - \frac{\lambda}{2} \quad \text{if} \quad \hat\beta_{k,\mathrm{LS}} > \frac{\lambda}{2} \qquad (\beta_k > 0).}$$

Case $\beta_k < 0$

Similarly one can show that

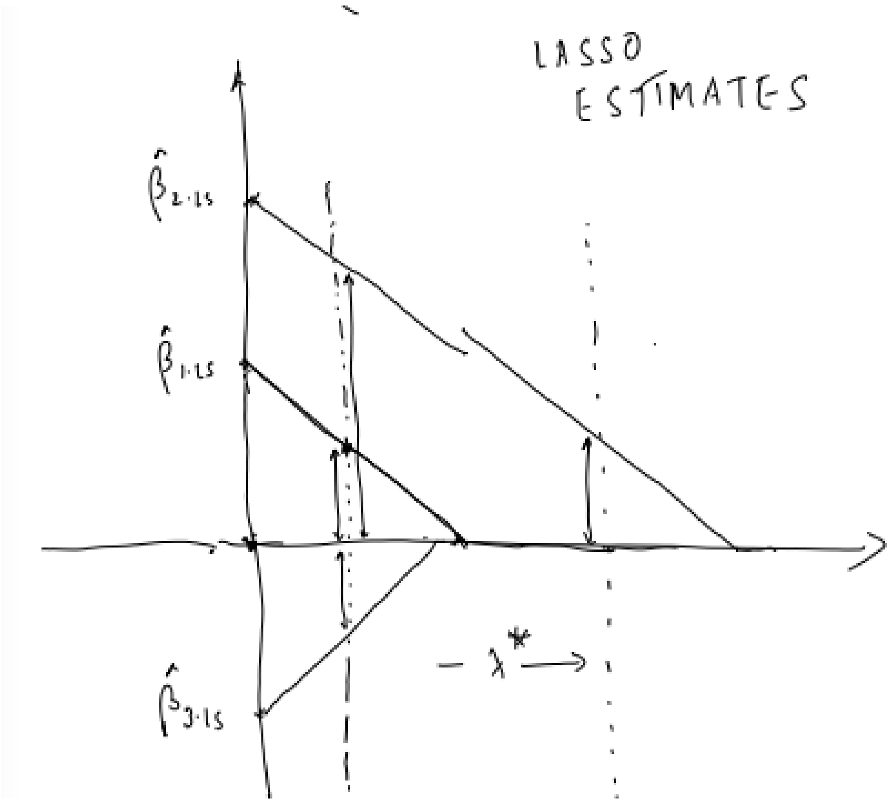
$$\boxed{\hat\beta_{k,\mathrm{LASSO}} = \hat\beta_{k,\mathrm{LS}} + \frac{\lambda}{2} \quad \text{if} \quad \hat\beta_{k,\mathrm{LS}} < -\frac{\lambda}{2} \qquad (\beta_k < 0).}$$

Case $\beta_k = 0$

What if $\beta_k = 0$? The condition $0 \in \partial\Delta(\beta_k)$ becomes

$$\hat\beta_{k,\mathrm{LS}} + c\frac{\lambda}{2} = 0 \quad\Rightarrow\quad \big|\hat\beta_{k,\mathrm{LS}}\big| \le \frac{\lambda}{2}.$$

$$\begin{aligned}\partial_0(\beta_k) &= \sum_{i=1}^{n}\left(y_i - \sum_{j=1}^{p}\beta_j x_{ji}\right)(-x_{ki}) + c\lambda \quad \text{where } c\in[-1,1]\\ &= -2\hat\beta_{k,LS} + 2\beta_k + c\lambda 0 \in \partial_0(\beta_k)\\ &\Rightarrow \hat\beta_{k,LS} + [-\frac{\lambda}{2}, \frac{\lambda}{2}] \ni 0 \Rightarrow |\hat\beta_{k,LS}| \le \frac{\lambda}{2}\end{aligned}$$



Soft-thresholding rule

Collecting the three cases we obtain the soft-thresholding estimator

$$\hat{\beta}_{k,\text{LASSO}} = \begin{cases} \hat{\beta}_{k,\text{LS}} - \lambda^* & \text{if } \hat{\beta}_{k,\text{LS}} > \lambda^*, \\ \hat{\beta}_{k,\text{LS}} + \lambda^* & \text{if } \hat{\beta}_{k,\text{LS}} < -\lambda^*, \\ 0 & \text{if } \left|\hat{\beta}_{k,\text{LS}}\right| < \lambda^*, \end{cases} \qquad \lambda^* = \frac{\lambda}{2}.$$

LASSO estimates

The objective function

$$\sum_{i=1}^{n}\Big(y_i - \sum_{j=1}^{p} \beta_j x_{ij}\Big)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

is the objective function of LASSO.

The value of λ can be chosen using cross-validation (CV).

So when $X^\top X = I$, the use of the penalty parameter λ gradually shrinks the magnitudes of the regression coefficients and eventually makes them zero (deletes the variable from the model) for higher values of λ.

How to compute the LASSO estimates if $X^\top X \neq I$?

We use the **coordinate descent** method for minimisation of the objective function. This is an iterative algorithm. We start with initial estimates $\beta_1^{(0)}, \beta_2^{(0)}, \ldots, \beta_p^{(0)}$ and update them iteratively.

$$X = \begin{bmatrix} x_{11} & x_{21} & \cdots & x_{p1} \\ x_{12} & x_{22} & \cdots & x_{p2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \cdots & x_{pn} \end{bmatrix}$$

$$\Delta = \sum_{i=1}^{n} \Big(y_i - \sum_{j=1}^{p} \beta_j x_{ij} \Big)^2 + 2\lambda \sum_{j=1}^{p} |\beta_j|.$$

In order to update the LASSO estimate for β_1 we minimise

$$\sum_{i=1}^{n} \Big(y_i - \beta_1 x_{1i} - \sum_{j=2}^{p} \beta_j^{(0)} x_{ji} \Big)^2 + 2\lambda \Big(|\beta_1| + \sum_{j=2}^{p} |\beta_j^{(0)}| \Big).$$

Define the partial residual

$$\sum_{i=1}^{n} 2(z_i - \beta_1 x_{1i})(-x_{i1}) = -2 \sum_{i=1}^{n} z_i x_{1i} + 2\beta_1 \sum_{i=1}^{n} x_{1i}^2$$

Then the problem reduces to

$$\sum_{i=1}^{n} (z_i - \beta_1 x_{i1})^2 + 2\lambda |\beta_1| \quad (+\lambda \text{const}), \text{where } z_i = y_i - \sum_{j \neq 1} \beta_j^{(0)} x_{ji}$$

The derivative with respect to β_1 (when $\beta_1 > 0$) is

$$-2 \sum_i z_i x_{i1} + 2\beta_1 \sum_i x_{i1}^2 + 2\lambda.$$

Equating it with zero one gets

$$\beta_1^{(1)} = \frac{\sum z_i x_{i1} - \lambda}{\sum x_{i1}^2} \quad \text{if} \quad \sum z_i x_{i1} > \lambda.$$

$$\beta_1 = \frac{\sum z_i x_{ii} - \lambda}{\sum x_{ii}^2} \quad \text{if} \quad \sum z_i x_{ii} > \lambda.$$

Similarly we can derive the expression for $\beta_1^{(1)}$ in the other two cases. Combining all three cases we obtain

$$\beta_1^{(1)} = \begin{cases} \dfrac{\sum z_i x_{i1} - \lambda}{\sum x_{i1}^2} & \text{if } \sum z_i x_{i1} > \lambda, \\ \dfrac{\sum z_i x_{1i} + \lambda}{\sum x_{1i}^2} & \text{if } \sum z_i x_{1i} < -\lambda, \\ 0 & \text{if } \left|\sum z_i x_{i1}\right| < \lambda. \end{cases}$$

When $X^\top X = I$ we have $\sum x_{i1}^2 = 1$ and

$$\sum z_i x_{1i} = \sum y_i x_{1i} - \sum_{j \neq 1} \beta_j^{(0)} \underbrace{\sum_i x_{ji} x_{1i}}_{=0} = \hat{\beta}_{1,\text{LS}},$$

recovering the soft-thresholding rule of the previous section.

This coordinate-descent method works as long as the non-differentiable part of the objective function is separable.

2 Nonparametric Regression

Here we want a piecewise linear estimate which is continuous. We divide the whole range of x into several parts

$$(-\infty, t_1),\ (t_1, t_2),\ \ldots,\ (t_{k-1}, t_k),\ (t_k, +\infty)$$

and look for a function which is

i) linear on each (t_{j-1}, t_j),

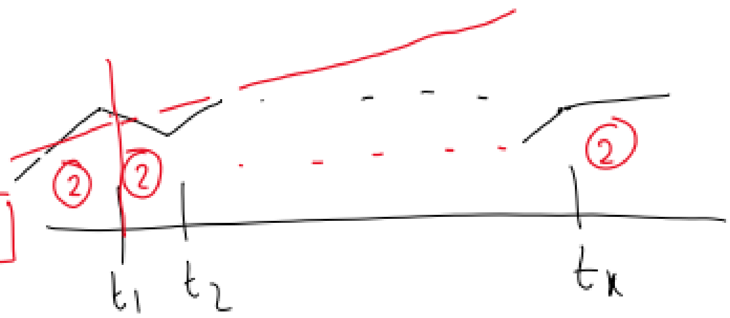
ii) continuous at the knots $t_1, \ldots, t_k$.

Such a function is called a **linear spline**.

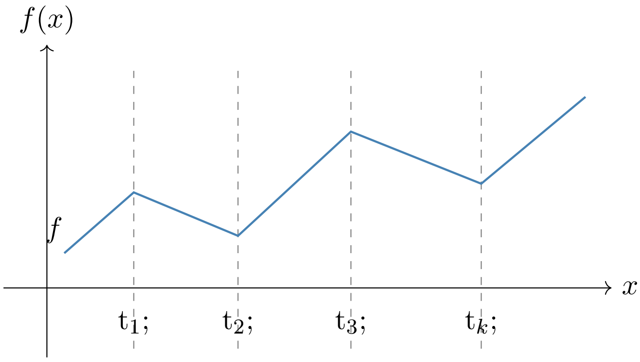
$2(k+1)$

$-k$

$\boxed{= k+2}$



$1,\ x,\ (x-t_1)_+,\ (x-t_2)_+,\ \ldots\ (x-t_k)_+$



A basis for linear splines is given by the truncated-power functions

$$1, \quad x, \quad (x - t_1)_+, \quad (x - t_2)_+, \quad \ldots, \quad (x - t_k)_+,$$

where the positive-part function is defined by

$$(x - t)_+ = \begin{cases} x - t & \text{if } x > t, \\ 0 & \text{if } o.w, \end{cases} \qquad \text{i.e.} \quad (x - t)_+ = \max\{0, x - t\}.$$

A general linear spline may therefore be written

$$f(x) = \beta_0 + \beta_1 x + \sum_{j=1}^{k} \beta_{j+1}(x - t_j)_+.$$

End of notes